Harshana Lakshara Fernando

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Summary

aBackend-focused Computer Science Engineering undergraduate with a rigorous foundation in **System Design**, **Distributed Systems**, and **Cloud-Native Engineering**. Proven experience architecting high-performance, fault-tolerant applications using **Microservices**, **Serverless Architectures**, and **Event-Driven Design** with **Java (Spring Boot)**, **Go**, **AWS Lambda**, and **Kafka**. Validated through the delivery of high-throughput systems handling massive load, seeking to solve complex **concurrency** and **low-latency** challenges within high-velocity engineering teams.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering *2022 – 2026*
University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream) *2019*
St. John's College, Panadura, Sri Lanka
Results: 3 A's (Combined Mathematics, Physics, Chemistry) **District Rank:** 17

Experience

Software Engineer — Trading Systems *Sep 2026 – Present*
CAL | Colombo, Sri Lanka

- Sole engineer on CAL's in-house electronic trading platform: a direct gateway to the **Colombo Stock Exchange (CSE)** plus the trading screens and ops tools around it, so the firm owns its platform instead of a vendor-built broker system.
- Took over **CSE certification** part-way and carried it to the final stage: every application-level case of the **188** exchange test cases re-run on CSE's test system with a full exchange-message log; only failover and trade-bust cases that CSE must trigger remain.
- Built an **event-sourced, CQRS** engine: a **Chronicle Queue** write-ahead log is the source of truth on the order path, with **PostgreSQL** as an async read model, so database writes never block execution.
- Implemented **FIX 5.0 SP2 / FIXT 1.1** order routing and drop copy (**QuickFIX/J**), a custom **Java NIO** decoder for the **MITCH** UDP market-data feed, and client-side token-bucket rate limiting.
- Built and own the Certification Runner (matches each test case to live exchange messages and explains every field difference from CSE's expected result), an order-lifecycle view (New, Fill, Cancel, Amend, Reject, Bust), the ops console, Pro dealing ticket and CAL Trader, plus durable message logs and an audit-ready test register.
- Using AI-assisted engineering (Claude), one person covers what is normally a team's work; design and code passed two senior reviews. Next: live CSE demonstration and sign-off, then Phase 2 (zero-allocation parsing with **Agrona**, active-passive standby).

R&D Software Engineer — KōreroCare Platform *Mar 2026 – Aug 2026*
Empathic Computing Laboratory, University of Auckland | Remote / Part-time

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Co-founder & Lead Engineer (Freelance) *Jul 2025 – Aug 2026*
ScaleX — SaaS Development Lab *Remote / Sri Lanka*

- **Stream-Native Tipping Platform:** Architecting a creator-focused SaaS to solve Sri Lankan payout barriers by integrating PayHere's aggregated gateway. Developed an OBS-native alert system using webhook-driven `notify_url` callbacks to trigger real-time on-stream overlays.

- **High-Performance Scheduling SaaS:** Engineering a mobile-first scheduling tool using Next.js 14 and Supabase Realtime for live heatmap synchronization. Implemented a custom best-slot recommendation algorithm to rank optimal meeting times based on participant overlap and duration constraints.
- **Scalable Infrastructure & DevOps:** Designed a $0-to-launch deployment strategy leveraging Vercel serverless functions and Supabase PostgreSQL with connection pooling. Established robust security protocols including Zod-based input validation and rate-limiting to ensure system resilience.
- **Strategic Compliance & Growth:** Navigating CBSL regulatory frameworks to operate as a software platform while delegating fund-handling to licensed gateways to minimize operational risk. Implementing a "freemium" viral loop strategy with branded footers and Stripe-integrated Pro tiers for monetization.
- **AgriNova:** Built a full-stack inventory and commission management platform (**Next.js**, **Prisma**, **Supabase**/**PostgreSQL**) replacing spreadsheet workflows: 5 branches, 329+ SKUs, 194K+ inventory units tracked and 860+ invoices processed (Rs. 64.4M+) with automated commission calculations. Cut repeat dashboard data-fetch latency from ∼384ms to under 1ms with React Query caching and pagination.

Software Engineer Intern — Information Systems Associates (Pvt) Ltd *Dec 2024 – Jun 2025*
Aviation Technology | Colombo, Sri Lanka

- **AeroOrder — Distributed PNR Generation Service**
  - Owned design and delivery of a distributed **Spring Boot**/**Kafka**-backed PNR generation service, **live in production**, achieving **25+ PNRs/sec** at **<100ms latency**.
  - Resolved production **Kafka consumer lag** and offset reset issues using shutdown hooks, retry handling, validation, and a Producer-Consumer refactor.
  - Led the legacy-to-microservice PNR rollout using database-driven **feature toggles** and a **strangler-pattern** migration approach.
  - Built load-testing and monitoring support using **JMeter**, **Prometheus**, and **Grafana** to validate async order/PNR flows, queue time, and service latency.
  - Maintained **>90% unit test coverage** while coordinating delivery with QA and distributed engineering teams.
- **AeroMart — Legacy Reservation System Integration**
  - Integrated **gRPC** clients and **Kafka** consumers into the **Java 8** AeroMart monolith for backward-compatible communication with AeroOrder services.
- **Engineering Practices**
  - Authored comprehensive design docs, sequence diagrams, and integration plans; facilitated knowledge transfer sessions with distributed engineering teams.
  - Added **Jenkins** environment support and streamlined service logging by replacing heavyweight logging dependencies with a lightweight structured JSON logging approach.

Open Source Contributions — See all

- **kgateway #14851** (CNCF) *Merged, Oct 2026*
  Added the **kgateway_build_info** Prometheus metric (version, revision, Go version, OS/arch) so operators can see which build each pod runs; shipped in **kgateway v2.5.0-beta.2**.
- **kubernetes-mcp-server #1482** *Merged, Oct 2026*
  Fixed flaky HTTP server startup in the **Go** test suite with a 5 s readiness poll and cleanup before failing.
- **kubernetes-mcp-server #1549** *Merged, Oct 2026*
  Test that every tool, prompt and resource across all 12 toolsets declares valid **Kubernetes RBAC** metadata; approved on the first review.
- **Kubeflow Spark Operator #3249** *In review (LGTM)*
  20 **Ginkgo** specs covering every executor pod-phase transition: **100%** of `sparkpod_metrics.go`, plus a log message fix.
- **KEDA #8317** (CNCF) *In review*
  New **onlyReadyPods** option for the Kubernetes workload scaler (count only Running and Ready pods); design agreed with a maintainer, docs in keda-docs #1919.
- **kubernetes-mcp-server #1568** *Root-caused, Oct 2026*
  Showed a reported "flaky" test was a real merge-order bug; the author confirmed and applied the suggested fix.

Projects — See more projects on my portfolio

1. Improving Blockchain Scalability: Throughput Enhancement in ZK-Rollups

Research Project — University of Moratuwa

- Built a modular ZK-rollup benchmarking framework to evaluate Ethereum Layer-2 trade-offs across sequencing, batching, proving, data availability, and L1 settlement.
- Implemented the Rust-based Submitter service for finalized batch submission, DA-mode selection, L1 bridge interaction, transaction receipt validation, and settlement metric export.
- Added calldata, EIP-4844-style blob, and off-chain / Validium-style benchmark modes, observing approximately 70% lower DA publication cost for blob-style mode compared with calldata.
- Introduced a mock-testable `BridgeClient` abstraction, improving Submitter testability and increasing verifiable test coverage from approximately 41% to 88%.
- Built Docker-based benchmark orchestration scripts, a Poisson-style workload generator supporting 8 TPS base and 80 TPS burst workloads, and Python data tools for throughput, P50/P95/P99 latency, finalization time, gas/cost, and DA-footprint analysis.
- Produced reproducible benchmark outputs including latency CDFs, cost heatmaps, and Pareto-style trade-off comparisons for the final research evaluation.

Technologies: Rust, Solidity, Hardhat, TypeScript, Python, Docker, gRPC, SQLite/PostgreSQL, RISC0 / Mock Proving Path, Matplotlib

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2. Blockchain Supply Chain Management DApp (2025 | Individual) Architecture | Live | Github

Built a full-stack decentralized application (DApp) enabling end-to-end shipment tracking and secure escrow payments on the Ethereum blockchain (Polygon Amoy Testnet).

- Developed a production-grade Solidity smart contract with a strict 3-state lifecycle (PENDING → IN_TRANSIT → DELIVERED) and on-chain auditability.
- Implemented a trustless escrow mechanism that locks funds on creation and releases payments only upon verified delivery.
- Built a responsive Next.js/React frontend with global state management and Ethers.js integration for secure wallet connections.
- Designed a CI/CD pipeline using GitHub Actions to automate contract and frontend deployment, utilizing Hardhat for local simulation and verification.

Keywords: Solidity, Hardhat, Ethers.js, Web3Modal, Next.js, React Context API, Tailwind CSS, Polygon Amoy, Smart Contracts, Escrow Payments, CI/CD

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3. Governance DAO – Token-Weighted Voting DApp (2025 | Individual)

Developed a decentralized governance platform with **token-weighted voting** using Solidity smart contracts.

- Implemented **GovToken** (ERC-20) using OpenZeppelin Contracts, with faucet-based token claims, owner-controlled minting, and airdrops.
- Built **VotingDAO** smart contract enabling proposal creation, weighted voting, quorum enforcement, and post-deadline execution.
- Deployed and tested contracts on Ethereum network using **Hardhat**.

Keywords: Ethereum, Solidity, Web3.js, Hardhat, OpenZeppelin

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4. Fintech & Crypto Trading Automation Lab (Ongoing | Group)

Industry-Mentored Initiative (Ongoing)

- Selected for an exclusive initiative to build intelligent automation systems for fintech and crypto markets under industrial mentorship.
- Developing production-ready trading tools and gaining deep exposure to technical analysis (TA) and algorithmic trading strategies.
- Focusing on high-performance software engineering with strict deadlines and client-side expectations.

Keywords: Fintech, Crypto Trading, Forex, Algorithmic Trading, Technical Analysis, Intelligent Automation, Industrial Mentorship, Personal Finance & Wealth Management

5. AgriNova Inventory & Commission Platform (2025 | Freelance, ScaleX) Live demo

Designed and delivered a production-grade inventory and sales management system for a real-world client, supporting multi-branch stock operations, invoicing, receipts, and commission tracking.

- Engineered transaction-safe stock movement logic and replaced application-layer aggregation with **PostgreSQL** aggregations, reducing execution time from **∼384 ms to ∼0.02 ms**.
- Designed a multi-layer caching strategy with **Next.js** and **React Query**, reducing repeated dashboard latency from **∼703 ms to ∼0.05 ms**.

- Built a clean, dynamic admin-focused UI and validated system reliability through end-to-end testing with **Playwright**.

Keywords: Next.js, PostgreSQL, Prisma, React Query, Tailwind CSS, Playwright

Training & Certifications

CKA Training — KodeKloud, 2025 • AWS SAA-C03 Training — Udemy, 2025 • Apache Kafka Series — Udemy, 2025 • 100 Days of DevOps — KodeKloud, 2025 • Advanced OAuth Security — Udemy, 2024

Skills

Core	**Java** | **Spring Boot** | **Microservices** | Distributed Systems | System Design | **REST APIs**
Backend	Spring Cloud | Spring Security | Resilience4j | Node.js | Express.js | Gin
Messaging & APIs	**Kafka** | RabbitMQ | **gRPC** | **GraphQL** | WebSockets | Protobuf | JWT | OAuth 2.0
Data	**PostgreSQL** | MySQL | Oracle DB | Redis | Couchbase
Cloud & DevOps	AWS | **Docker** | Kubernetes | CI/CD (GitHub Actions, Jenkins)
Testing	JMeter | Gatling | Pytest | Unit Testing
Tools	Git | Gradle | Postman | Jira | Confluence

References available upon request.